

Context

Act as senior test engineer.

Instruction

[STRICT] Generate Functional test cases for the Login page with MFA. [MANDATE] Include Positive and Negative scenarios [MANDATE] Include Edge cases [MANDATE] Prioritize the test cases

Persona Act as Senior Manager to review test cases

Output Generate test cases in downloadable excel file

Tone Generate comments in professional tone

Chatgpt result:

I have prepared a comprehensive set of Functional Test Cases for the Login Page with MFA, including:

- Positive scenarios
- Negative scenarios
- Edge cases
- Priority classification (High / Medium / Low)
- Professional review comments suitable for Senior Manager review

Since downloadable Excel generation requires login-enabled tools in this environment, you can directly copy the CSV content below into Excel and save it as .xlsx.

TC_ID,Priority,Scenario Type,Test Scenario,Preconditions,Test Steps,Test Data,Expected Result,Review Comments

TC_LOGIN_001,High,Positive,Login with valid username/password and valid MFA OTP,User account active and MFA enabled,"1. Navigate to login page

2. Enter valid username and password

3. Click Login

4. Enter valid OTP

5. Click Verify", "Valid credentials, valid OTP", User should successfully login and land on dashboard, Core business flow should always pass in regression suite.

TC_LOGIN_002, High, Negative, Login with invalid password, User account active and MFA enabled, "1. Navigate to login page

2. Enter valid username and invalid password

3. Click Login", "Valid username, invalid password", System should display appropriate error message and deny access, Validate error message does not expose sensitive information.

TC_LOGIN_003, High, Negative, Login with invalid username, User account active and MFA enabled, "1. Navigate to login page

2. Enter invalid username and password

3. Click Login", Invalid username, System should deny access and display generic authentication error, Prevent username enumeration.

TC_LOGIN_004, High, Negative, Login with valid credentials but invalid MFA OTP, User account active and MFA enabled, "1. Enter valid credentials

2. Enter invalid OTP

3. Click Verify", Invalid OTP, System should reject OTP and remain on MFA page, Critical security validation.

TC_LOGIN_005, Medium, Positive, Resend MFA OTP and login successfully, User account active and MFA enabled, "1. Login with valid credentials

2. Click Resend OTP

3. Enter newly received OTP", New OTP, User should login successfully using latest OTP, Validate previous OTP invalidation.

TC_LOGIN_006, High, Negative, Attempt login with expired MFA OTP, User account active and MFA enabled, "1. Login with valid credentials

2. Wait until OTP expires

3. Enter expired OTP",Expired OTP,System should reject expired OTP and prompt resend option,Important for session security.

TC_LOGIN_007,Medium,Edge Case,Verify OTP field boundary validation,User account active and MFA enabled,"1. Login with valid credentials

2. Enter OTP with less/more than allowed digits","OTP lengths: 0, 1, max+1",System should validate OTP length correctly,Boundary validation should be automated.

TC_LOGIN_008,Medium,Negative,Account lock after consecutive failed password attempts,User account active and MFA enabled,1. Enter invalid password repeatedly until threshold reached,Multiple invalid passwords,Account should lock after configured attempts,Ensure audit logs are generated.

TC_LOGIN_009,Medium,Negative,Account lock after consecutive failed OTP attempts,User account active and MFA enabled,"1. Login with valid credentials

2. Enter invalid OTP repeatedly",Multiple invalid OTPs,System should temporarily lock MFA verification,Protects against brute-force attacks.

TC_LOGIN_010,Low,Positive,Remember me functionality after successful MFA login,User account active and MFA enabled,"1. Select Remember Me

2. Login successfully with MFA

3. Relaunch browser",Valid credentials,Application should maintain session as per policy,Validate session timeout configuration.

TC_LOGIN_011,Medium,Edge Case,Verify login with leading/trailing spaces in username,User account active and MFA enabled,"1. Enter username with spaces

2. Enter valid password",Username with spaces,System should trim spaces or display validation message,Avoid authentication inconsistencies.

TC_LOGIN_012,High,Negative,SQL injection attempt in username field,Application accessible,"1. Enter SQL payload in username

2. Attempt login"," OR '1'='1",Application should sanitize input and deny access,Security test mandatory before production release.

TC_LOGIN_013,High,Negative,Cross-site scripting attempt in login fields,Application accessible,1. Enter script payload in username/password fields,<script>alert(1)</script>,Application should sanitize script input,Validate secure coding controls.

TC_LOGIN_014,Medium,Positive,Login from supported browser,Supported browser installed,"1. Open application in supported browser

2. Perform successful login",Chrome/Edge/Firefox,User should login successfully,Cross-browser coverage required.

TC_LOGIN_015,Low,Edge Case,Verify behavior on browser refresh during MFA step,User at MFA page,"1. Login with valid credentials

2. Refresh browser during MFA step",N/A,Session should remain stable or redirect safely,Ensure no duplicate OTP generation.

TC_LOGIN_016,Medium,Negative,Attempt reuse of already used OTP,User account active and MFA enabled,"1. Login successfully

2. Logout

3. Attempt reuse of previous OTP",Previously used OTP,System should reject reused OTP,OTP replay prevention required.

TC_LOGIN_017,High,Positive,Login with backup recovery code,Recovery codes generated,"1. Login with valid credentials

2. Use recovery code instead of OTP",Valid recovery code,User should login successfully,Validate single-use behavior separately.

TC_LOGIN_018,Medium,Negative,Login with invalid recovery code,Recovery codes generated,"1. Login with valid credentials

2. Enter invalid recovery code",Invalid recovery code,System should deny access,Recovery flow should follow same security standards.

TC_LOGIN_019,Low,Edge Case,Session timeout during MFA verification,User at MFA page,"1. Login with valid credentials

2. Remain idle until timeout

3. Enter OTP",Valid OTP after timeout,System should expire session and request re-login,Important usability and security balance.

TC_LOGIN_020,High,Positive,Verify audit logging for successful and failed login attempts,Audit logging enabled,1. Perform successful and failed login attempts,"Valid and invalid credentials",System should capture audit events accurately,Compliance-related validation.

TL;DR

MFA Login Test Cases — Summary

- Total Test Cases: **20**
- Coverage Includes:
 -  Positive Scenarios
 -  Negative Scenarios
 -  Edge Cases
 -  Security Validations
 -  Session & OTP Handling
 -  Recovery & Lockout Flows

Priority Distribution

-  High Priority: Core authentication & security flows
-  Medium Priority: Validation & usability checks
-  Low Priority: Session behavior & browser handling

Key Areas Covered

- Valid/Invalid login
- MFA OTP verification
- Expired/reused OTP
- Account lockout
- Recovery code flow
- SQL Injection & XSS
- Session timeout
- Browser refresh handling
- Audit logging

Deliverable

- Structured in Excel-ready CSV format
- Professional review comments included

- Suitable for QA review, regression, and audit preparation

ELI-15

What is Transformers in AI

Imagine you are in a classroom and the teacher asks:

“Who in this paragraph is ‘he’ referring to?”

Your brain quickly looks at the important words nearby and understands the meaning.

A Transformer in AI works almost the same way.

Super Simple Explanation

A Transformer is a smart AI brain that learns:

- which words are important,
- how words connect,
- and what the full meaning is.

Instead of reading one word slowly at a time, it looks at the whole sentence together and focuses on the important parts.

That focusing ability is called:

“Attention”

Example:

“Ravi dropped the glass because he was careless.”

The Transformer understands:

- “he” = Ravi
- not the glass

Because it pays attention to relationships between words.

Why Is It Powerful?

Older AI models had memory problems with long sentences.

Transformers:

- understand long conversations better,
- work much faster,
- and can learn from huge internet-scale data.

That's why modern AI became so advanced after Transformers were invented.

Real-Life Apps Using Transformers

Things you already use:

- ChatGPT
- Google Translate
- AI coding assistants
- Voice assistants
- AI image tools

All mostly run on Transformer technology.

Easy Analogy

Think of Transformers like:

A super-intelligent student who:

- reads very fast,

- remembers context,
- connects ideas,
- predicts the next word intelligently.

Example:

You type:

“Write login test cases”

Transformer AI thinks:

- User wants testing content
- Maybe functional testing
- Include positive/negative cases
- Generate structured response

Then it creates the answer.

One-Line ELI-15 Definition

Transformers are AI systems that learn meaning by paying attention to important relationships in data, especially words and context.